

2023 – 2024 Final Essay

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Question (1) a. Create pseudo code for Bresenham algorithm.

Bresenham algorithm :-

// Assume $m < 1$, $x_0 < x_1$

Step1 : input two line end points

Step2 : calculate initial values

$$\Delta x = x_1 - x_0$$

$$\Delta y = y_1 - y_0$$

$$P = 2\Delta y - \Delta x \quad y = y_0$$

$$x = x_0$$

WritePixel (x, y)

Step3 :

while ($x < x_1$) {

$$x = x + 1$$

if ($P < 0$)

{

$$P = P + 2\Delta y$$

} **else** {

$$P = P + 2(\Delta y - \Delta x)$$

$$y = y + 1$$

}

WritePixel (x, y)

}

Step4 : Repeat step3 dx steps.

Question (1) b. Trace and draw the steps of this algorithm to display the line between the two points (20,41) and (30,44) (10 Marks)

$$\Delta x = 30 - 20 = 10$$

$$\Delta y = 44 - 41 = 3$$

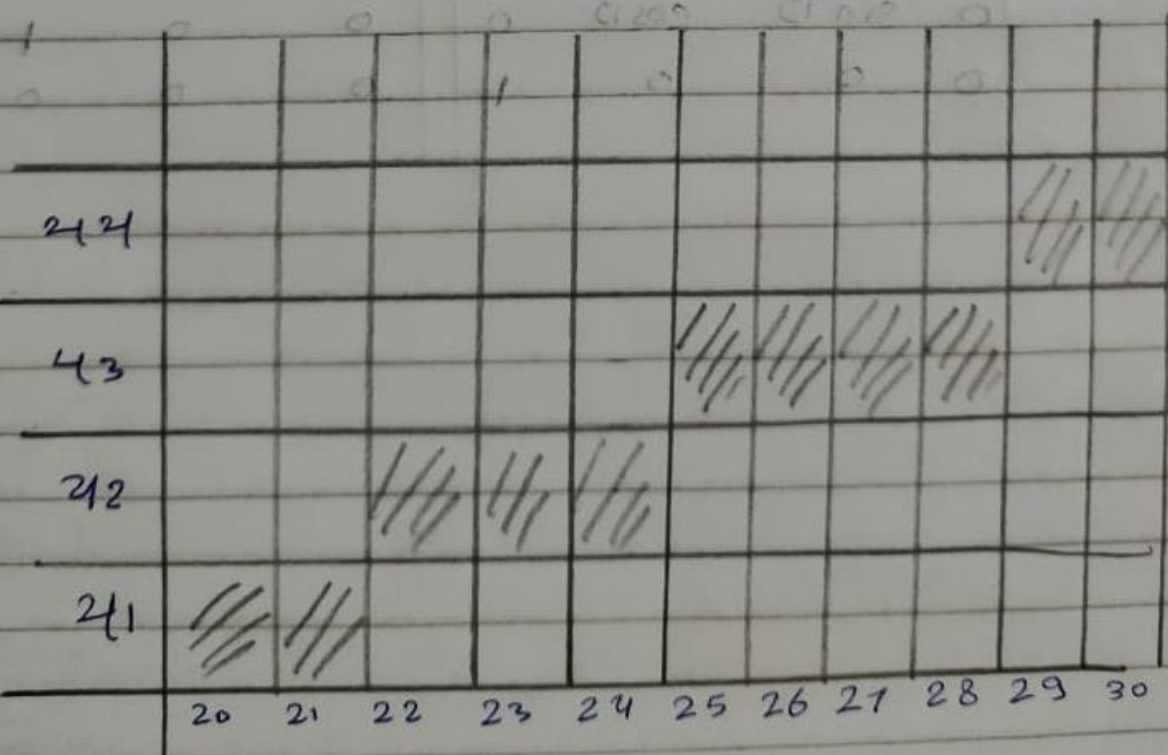
$$2\Delta x = 20$$

$$2\Delta y = 6$$

$$P = 6 - 10 = -4$$

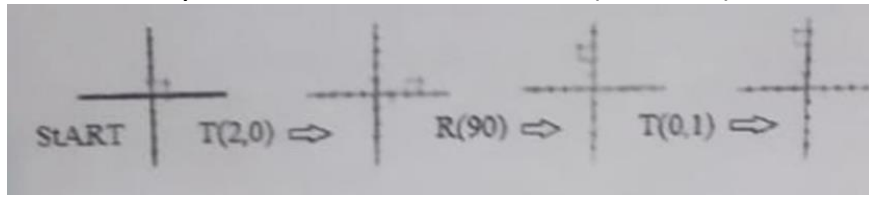
$$2\Delta y - 2\Delta x = 6 - 20 = -14$$

K	P_k	Pixel
0	-4	(21, 41)
1	$-4 + 6 = 2$	(22, 42)
2	$2 + 6 - 20 = -12$	(23, 42)
3	$-12 + 6 = -6$	(24, 42)
4	$-6 + 6 = 0$	(25, 43)
5	$0 + (-14) = -14$	(26, 43)
6	$-14 + 6 = -8$	(27, 43)
7	$-8 + 6 = -2$	(28, 43)
8	$-2 + 6 = 4$	(29, 44)
9	$4 + (-14) = -10$	(30, 44)



Question (2) a. Suppose we have the following sequence of 2D transformations: $T(2,0)$ $R(90)$ $T(0,1)$.

Assume you have a unit square (extending from 0 to 1 in x and y) at the origin. Draw out the intermediate three positions of the square as each transformation is applied. Write the transformation matrix in 2D homogeneous coordinates for each position. Give the matrix representing the entire sequence of transformations. (9 Marks)



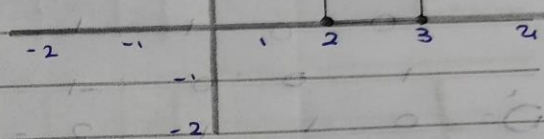
$$T(2,0)T_1 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 & 1 & 3 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 & 1 & 3 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$



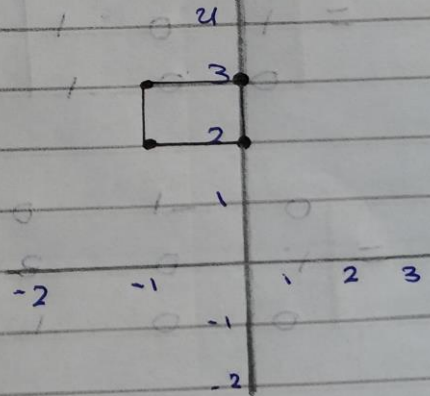
$$R(90)T_2 = \begin{bmatrix} \cos 90^\circ & -\sin 90^\circ & 0 & 0 & -1 & 0 \\ \sin 90^\circ & \cos 90^\circ & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & -1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & -1 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & 3 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & -1 & 0 & 3 & -1 \\ 1 & 0 & 0 & 1 & 3 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 & -1 & 0 & 2 & -1 \\ 1 & 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$



Solution Continued :

Date : _____
No: _____

$$T(0,1) \overline{I_3} = \begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array}$$

$$A' = \begin{array}{ccc|c|c|c} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 2 & = & 3 \\ 0 & 0 & 1 & 1 & & 1 \end{array}$$

$$B' = \begin{array}{ccc|c|c|c} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 3 & = & 4 \\ 0 & 0 & 1 & 1 & & 1 \end{array}$$

$$C' = \begin{array}{ccc|c|c|c} 1 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 1 & 3 & = & 4 \\ 0 & 0 & 1 & 1 & & 1 \end{array}$$

$$D' = \begin{array}{ccc|c|c|c} 1 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 1 & 2 & = & 3 \\ 0 & 0 & 1 & 1 & & 1 \end{array}$$

$$CT = \overline{I_3} \overline{I_2} \overline{I_1}$$

$$CT = \begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \end{array}$$

$$= \begin{array}{ccc|cc} 0 & -1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}$$

$$= \begin{array}{ccc|c} 0 & -1 & 0 & 3 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{array}$$

Question (2) b. In a 3D system a point is rotated by an angle Θ around the origin in x-y plane. Then the point is rotated by angle β around the origin in y-z plane. Derive the transformation matrix for this two step process. (6 Marks)

① Rotation by angle Θ in Z axis

$$T_1 = \begin{bmatrix} \cos \Theta & -\sin \Theta & 0 & 0 \\ \sin \Theta & \cos \Theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

② Rotation by angle β in X axis

$$T_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \beta & -\sin \beta & 0 \\ 0 & \sin \beta & \cos \beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$CT = T_2 T_1$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \beta & -\sin \beta & 0 \\ 0 & \sin \beta & \cos \beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \Theta & -\sin \Theta & 0 & 0 \\ \sin \Theta & \cos \Theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Question (3) c. What is the color model? State its types and explain RGB model.

Color Model :

- A color model is an orderly system for creating a whole range of colors from a small set of primary colors.

RGB Color Model : is an additive color model For computer displays uses light to display color , Colors result from transmitted light.

CMYK Color Model : is a subtractive color model and the standard color model used in offset printing for full-color documents. Because such printing uses inks of these four basic colors, it is often called four-color printing.

HSV Color Model : Hue, Saturation, Value or HSV is a color model that describes colors (hue or tint) in terms of their shade (saturation) and their brightness (value). HSV color model is based on polar coordinates.

Question (3) d. Explain how does LCD monitor display pictures?

Liquid Crystal Displays (LCDs) : It produces picture by passing polarized light through a liquid crystal material that can be aligned either block or transmit the light .

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